

SIH26249 — 01 Product Overview

SIH 2026 — SIH26249

Air Power: Predictive Maintenance & Fleet Availability

Ministry of Defence — Defence Services Staff College

01 — Product Overview: READYFLEET

Vision

An air force wins or loses on how many aircraft are mission-capable tomorrow morning — not on how many engines it owns. READYFLEET is the **fleet-availability decision layer**: it fuses component health predictions with spares, crew, technical records, and sortie demand into one forward forecast of mission-capable aircraft, and prescribes the maintenance action *today* that changes *tomorrow's* number.

The problem (PS SIH26249, verified 2026-10-03)

<https://sih2026-ps-viewer.vercel.app/ps/SIH26249>

"Low aircraft availability due to fragmented and largely reactive maintenance practices across the air fleet. Maintenance data from aircraft health-monitoring systems, technical records, spares and maintenance agencies is not adequately integrated, resulting in delayed fault prediction, avoidable aircraft downtime and sub-optimal utilisation of critical assets."

Two insights drive our design: 1. **Integration is the diagnosis, not prediction.** The PS's central complaint is that four data sources are "not adequately integrated." A better single-engine RUL model does not fix that. A fusion layer does. 2. **Availability is the decision variable.** Every maintenance system reports *component health*; nobody in the loop reports *how many aircraft fly tomorrow and what to do about it*. That gap is the product.

Personas

1. Base Commander (primary decision-maker)

- **Wants:** one number every morning — "how many tails are mission-capable today and through the week," plus "which 2–3 actions move it most."
- **Uses:** fleet board, MC forecast curve, NMC-driver ranking, what-if levers, daily readiness brief.
- **Success:** makes a defensible resourcing decision (expedite a part, authorize cannibalization, re-task crew) with the trade-off quantified before signing.

2. Maintenance Officer (the executor)

- **Wants:** which tails to work, in what order, with what parts and how many crew-hours — a prioritized, feasible work plan.
- **Uses:** tail drill-down, explainability panel, crew-capacity 1-day schedule, cannibalization advisor, discrepancy watchlist.
- **Success:** crew hours land on the highest mission-impact work; no tail sits waiting on a part nobody ordered.

3. Logistics / Spares Officer (the enabler)

- **Wants:** which parts will be needed *before* they are needed; which orders to expedite; which shortages ground aircraft.
- **Uses:** spares shortfall / MICAP-risk predictor, break-rate→parts-demand predictor, what-if lever "expedite spares."
- **Success:** expedite decisions backed by a measured MC delta, not a gut call.

PS requirement mapping

PS requirement	Feature that delivers it
Fault prediction earlier ("delayed fault prediction")	Per-component RUL with prediction intervals, 24 h critical-horizon alerts
Integrate health-monitoring data	HUMS-style health signal stream consumed as one of five inputs
Integrate technical records	Maintenance records feed discrepancy history, break-rate models, recurrence watchlist
Integrate spares data	Spares shortfall / MICAP-risk predictor; spares as a binding constraint in the aggregator
Integrate maintenance agencies / capacity	Crew-capacity constraint scheduler
Cut avoidable downtime	NMC-driver ranking: "fix X → recovers Y tails"
Better utilisation of critical assets	Mission-priority scoring; what-if levers maximize <i>mission-capable</i> hours

Success metrics (demo-day measurable)

1. **RUL model quality:** RMSE/MAE on held-out tails vs a naive mean-RUL baseline — printed by `eval.py`, stored in `metrics.json`. (Measured at demo; only claimed if the run says so.)
2. **What-if latency:** full scenario recompute < 200 ms (live in demo — the "before/after" number updates on click).
3. **Decision coverage:** every NMC tail in the demo scenario has a named binding constraint (NMCM/NMCS/NMCB) and a ranked driver.
4. **Honesty coverage:** automated test — every DB row has `synthetic=1`, every API response carries `"data_class": "synthetic"`, banner present on every screen.
5. **Demo delta:** scripted 7-day scenario shows a live-computed MC-rate improvement from a prescribed lever (e.g., expedite part → +N tails) — the number is computed in front of the judges, never pre-baked.

What READYFLEET is NOT

- Not a sensor box, not a competing HUMS — HAL's HUMS does per-platform diagnostics; we sit above it.
- Not a full MILP optimizer in 36 h — a greedy, guardrailed heuristic with the MILP path documented.
- Not a physics digital twin — our "twin" is a scenario twin (cloned fleet states for what-if replay), stated openly.
- Not real IAF data — everything synthetic, labeled, every screen, every API response.

Sources (plain full URLs)

<https://sih2026-ps-viewer.vercel.app/ps/SlH26249> <https://www.gao.gov/assets/t-gao-01-693.pdf>